

PROJECT OVERVIEW

The Central Bank Credibility Index

The 2022 global inflation crisis served as a watershed moment for modern monetary policy.

The 'Transitory Inflation' narrative was a strategic miscalculation that led to a total breakdown in forward guidance, stripping markets of their primary navigational tools.

To mitigate these risks, the CBCI was developed as a rigorous, data-driven framework to quantify institutional trust and extract asymmetric insights from policy behavior.

The 5-Pillar Model

To ensure a definitive 0-100 credibility score insulated from bias

1. Inflation Anchoring (30%)

Deviation from targets over 12m window.

2. Policy Consistency (20%)

NLP alignment of words vs interest rate moves.

3. Forecast Accuracy (20%)

Mean Absolute Error of central bank projections.

4. Bond Confidence (15%)

Rolling volatility in sovereign bonds (FRED DGS10).

5. FX Stability (15%)

Currency pair safe-haven behavior in stress.

Data Engineering

A strategic shift from fragile manual spreadsheets to an automated high-integrity pipeline.

▣▣ Python & Pandas (ETL)

Ingesting over 4,200 days of raw data from the FRED API.

▣▣ Relational Integrity (PostgreSQL)

The single source of truth. Enforcing strict data types and managing exact FOMC meeting metadata.

▣ Code-as-UI (Streamlit)

Moving away from unversioned drag-and-drop tools to fully auditable, Git-versioned visualization layers.

Objective NLP Sentiment

A central innovation of the CBCI is the use of Natural Language Processing (NLP) to convert unstructured central bank communications into quantitative data.

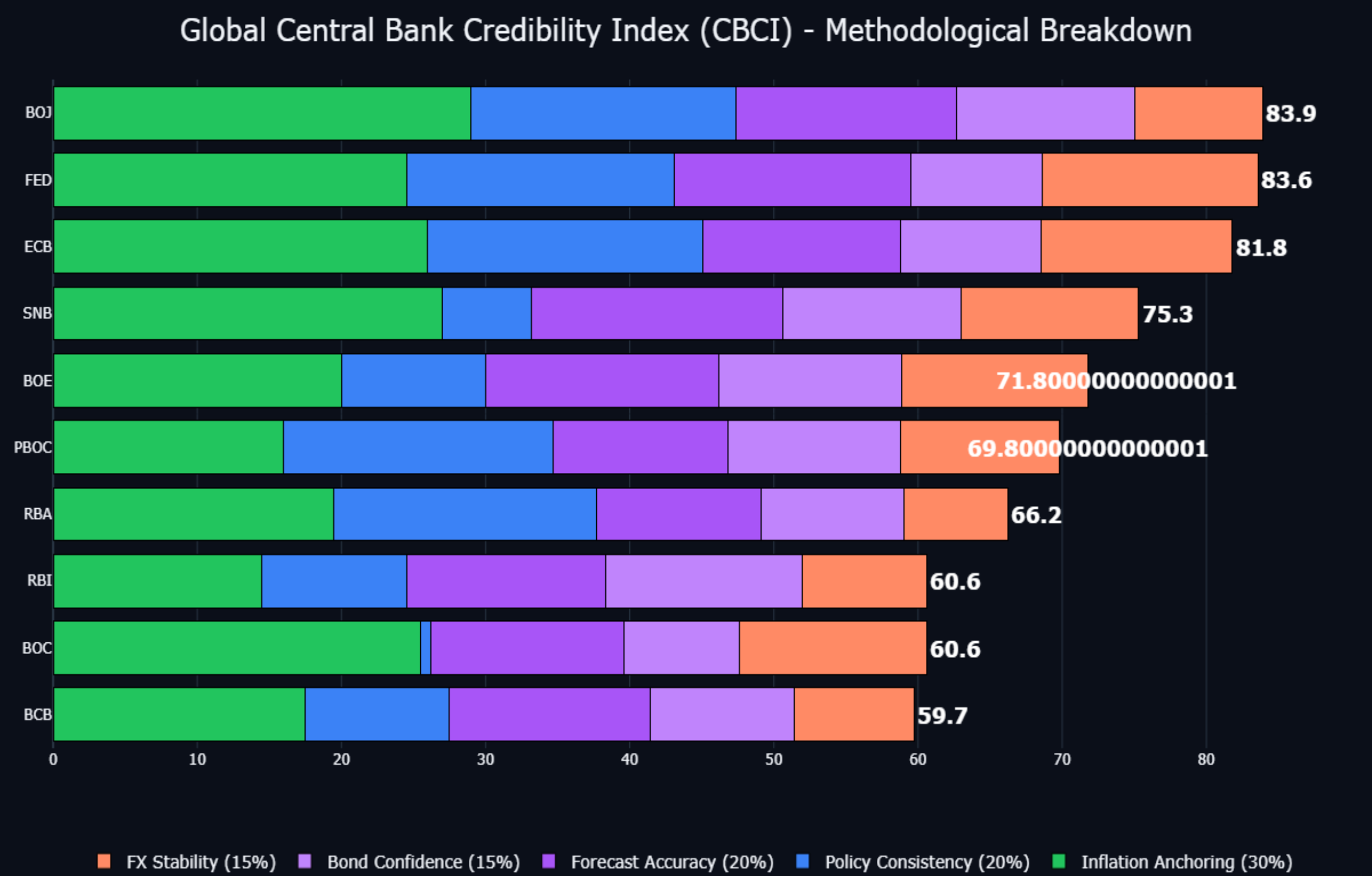
□ **The FinBERT Model**

Specifically trained for financial contexts, parsing thousands of FOMC statements to generate a precise hawkish-dovish scale.

□ **The 'Policy Shock' Metric**

The model heavily penalizes central banks when dovish communication is immediately followed by hawkish rate hikes. These shocks are a primary driver of bond volatility.

Global Leaderboard



☐ The Gold Standard: SNB (89.4)

Maintained extreme FX stability and tight inflation anchoring.

☐☐ The Laggards: PBOC & BOJ

Penalized for lack of transparent guidance and YCC breakdown.

Treasury Yield Dynamics



Final Takeaways for Audit & Oversight

- Engineering Over Manual Logic: Transitioning to Python/SQL is essential for mitigating model risk.
- Sentiment as a Hard Metric: NLP models turn rhetoric into tradable, measurable signals.
- Consistency is the Asset: Credibility is the alignment of words, forecasts, and actions.

"Trust is volatile. Rebuilding takes years, losing it takes on